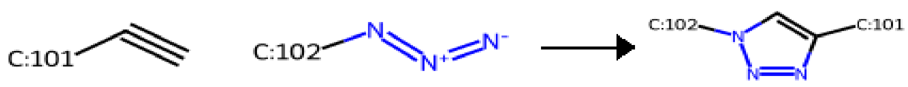
Reaction Catalog

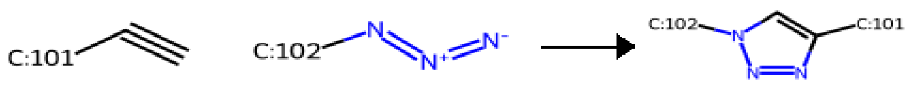
123-triazole-1

Copper-catalyzed Huisgen cycloaddition of an alkyne and an azide to form a 123-triazole (implicit azide formation from primary and secondary halides).



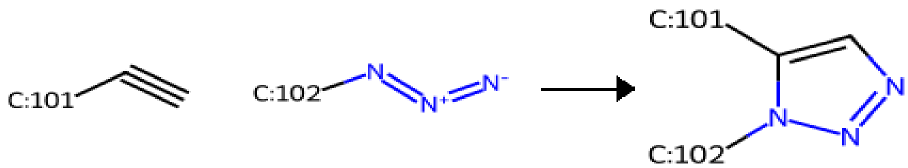
123-triazole-2

Copper-catalyzed Huisgen cycloaddition of an alkyne and an azide to form a 123-triazole.



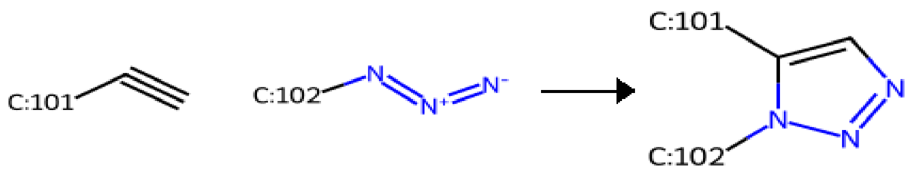
123-triazole-3

Ruthenium-catalyzed Huisgen cycloaddition of an alkyne and an azide to form a 123-triazole (implicit azide formation from primary and secondary halides).



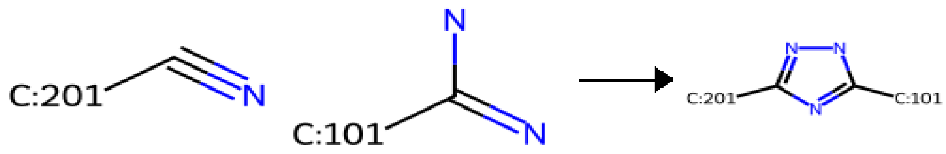
123-triazole-4

Ruthenium-catalyzed Huisgen cycloaddition of an alkyne and an azide to form a 123-triazole.



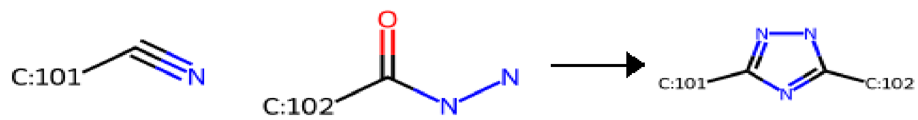
124-triazole-1

124-triazole from an aryl nitrile and an amidine.



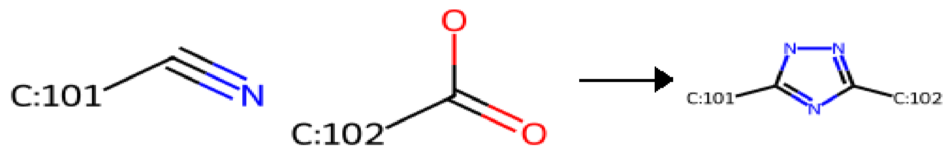
124-triazole-2

124-triazole from a nitrile and hydrazide.



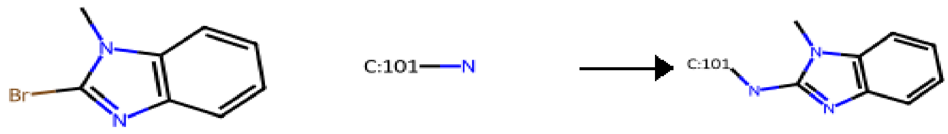
124-triazole-3

124-triazole from a nitrile and a carboxylate using hydrazine.



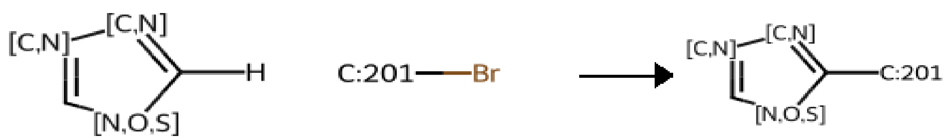
5-het-amination

Amination of 5-member heterocycles (n-alkyl-imidazole, thiazole, oxazole, isoxazole, benzothiazole, benzimidazole, and benzoxazole).



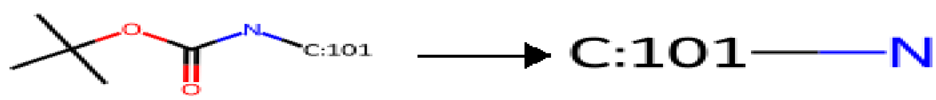
5-het-arylation

Arylation of 5-member heterocycles, including: thiophene, n-methyl-imidazole, 4-methyl-thiazole, 2-methyl-furan, benzothiophene, and benzothiazole.



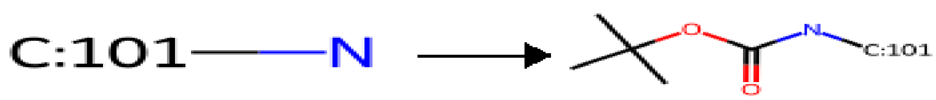
BOC-deprotection

Deprotection of a Boc-protected amine



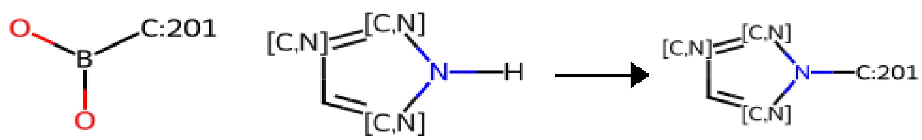
BOC-protection

Protection of a primary or secondary amine with a Boc-group



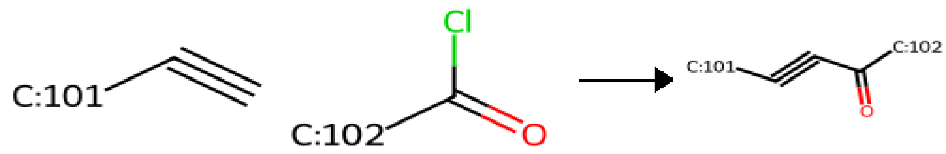
N-azole-arylation

N-arylation of azoles, including: indole, pyrazole, imidazole, triazole, tetrazole, benzimidazole, and indazoles.



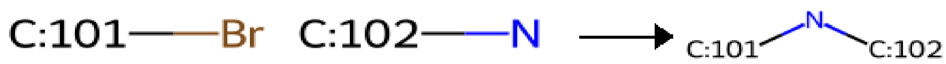
acylation-1

Acylation of an alkyne with an acid chloride.



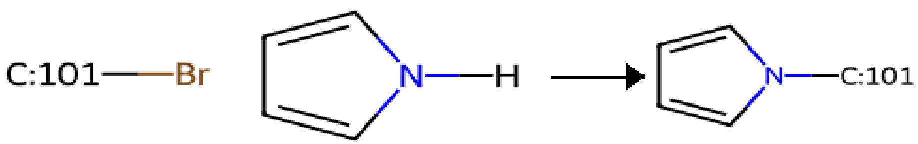
alkylation-1

Alkylation of a primary or secondary amine using a primary or secondary alkyl halide.



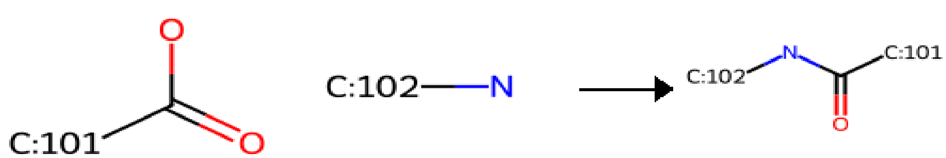
alkylation-2

N-Alkylation of heterocycles using a primary or secondary alkyl halide.



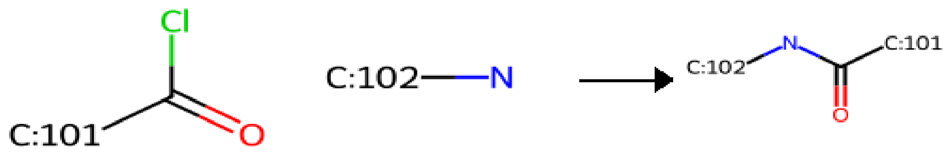
amide_coupling-1

Traditional amide coupling using a carboxylic acid and a primary or secondary amine.



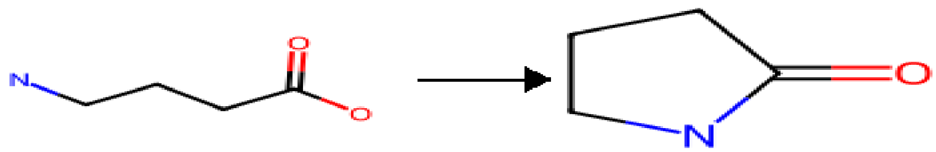
amide_coupling-2

Schotten-Baumann: Amide coupling using an acid chloride and a primary or secondary amine.



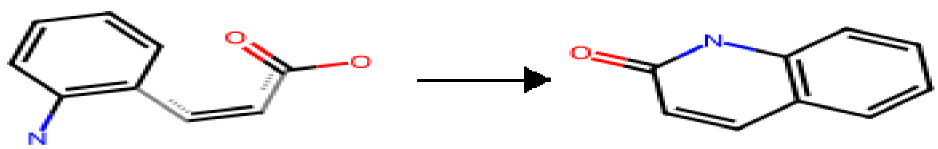
amide_coupling-3

Intramolecular amide coupling



amide_coupling-4

Intramolecular amide coupling

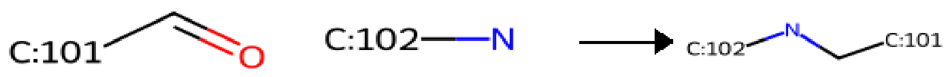


amination-1

Buchwald-Hartwig Amination: N-arylation of primary and secondary amines using aryl halides.

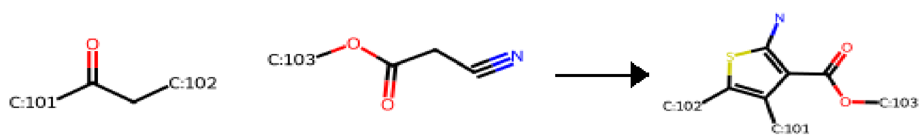
amination-2

Reductive Amination: Alkylation of primary or secondary amines with a ketone or an aldehyde.



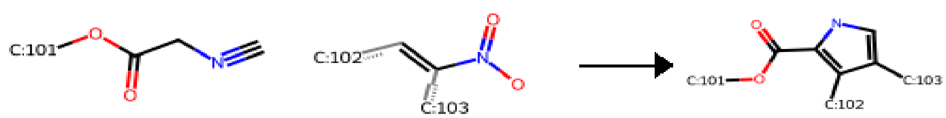
aminothiophene-1

Gewald Aminothiophene synthesis.



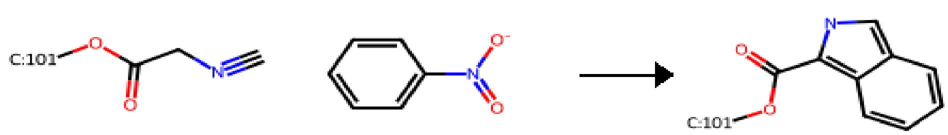
barton-zard-1

Synthesis of a substituted pyrrole from an isocyanoacetate and a nitroalkene



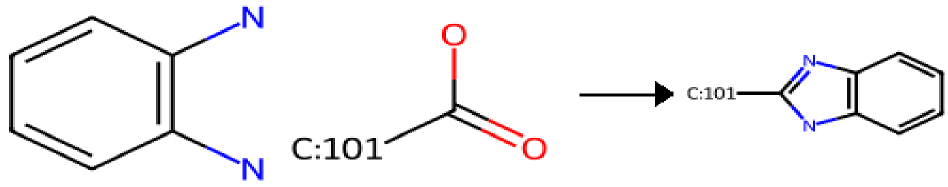
barton-zard-2

Synthesis of a substituted pyrrole from an isocyanoacetate and a nitrobenzene



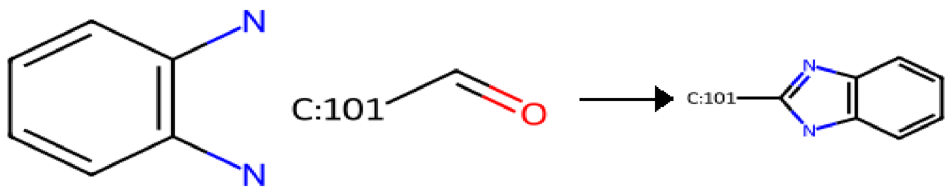
benzimidazole-1

Benzimidazole synthesis from an ortho-amino aniline and a carboxylic acid or ester.



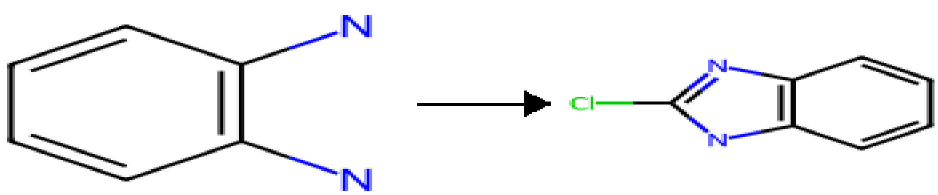
benzimidazole-2

Benzimidazole synthesis from an ortho-amino aniline and an aldehyde.



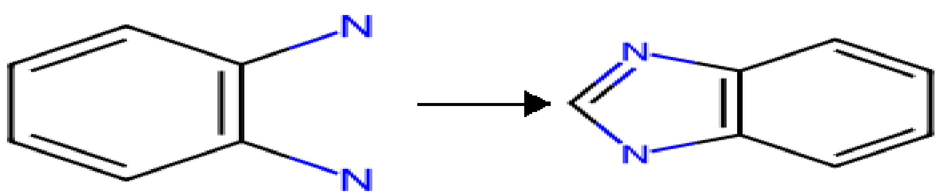
benzimidazole-3

2-Chloro-benzimidazole formation from 2-amino anilines.



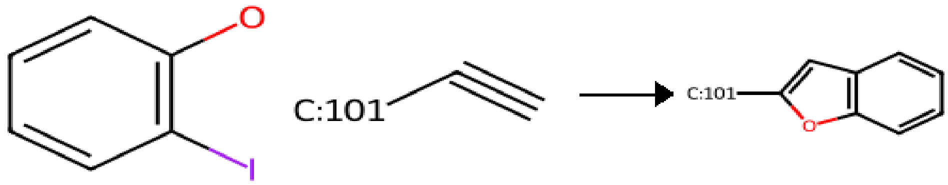
benzimidazole-4

Benzimidazole formation from 2-amino anilines.



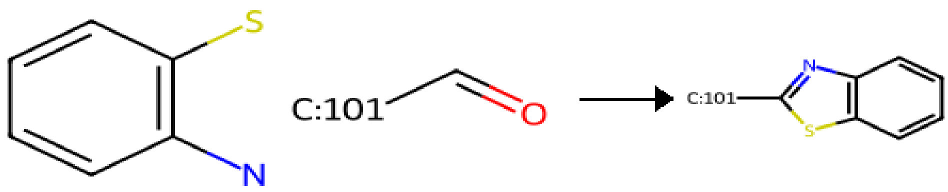
benzofuran-1

Benzofuran synthesis from a 2-iodo phenol and an alkyne.



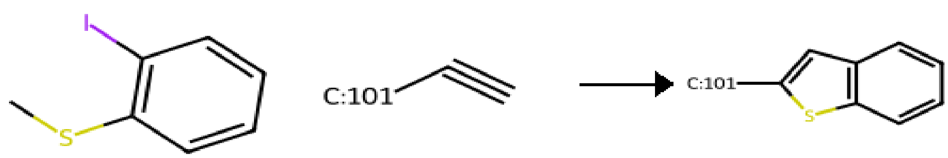
benzothiazole-1

Benzothiazole synthesis from a 2-thiol aniline and an aldehyde.



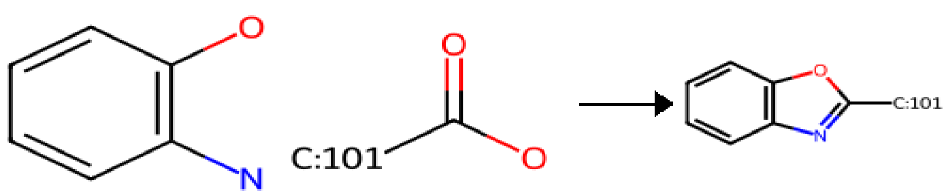
benzothiophene

Benzothiophene synthesis.



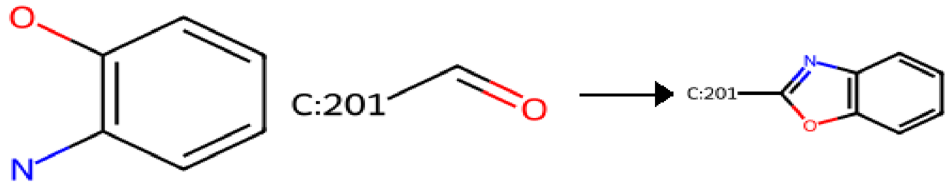
benzoxazole-1

Benzoxazole synthesis from a 2-amino phenol and a carboxylic acid.



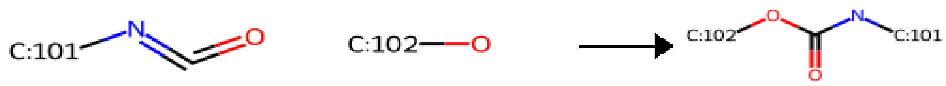
benzoxazole-2

Benzoxazole synthesis from a 2-amino phenol and an aryl aldehyde.



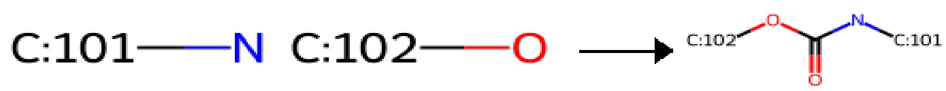
carbamate

Carbamate synthesis from isocyanates and primary or secondary alcohols.



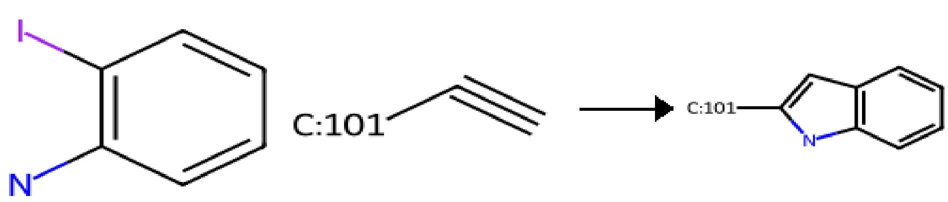
carbamate-2

Carbamate synthesis from primary or secondary amines and primary or secondary alcohols.



castro-stephens-1

Castro-Stephens coupling: synthesis of indoles.

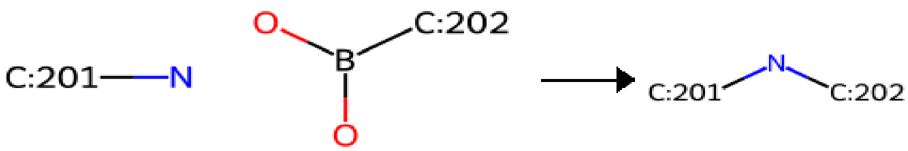


castro-stephens-2

Castro-Stephens coupling: synthesis of disubstituted acetylenes.

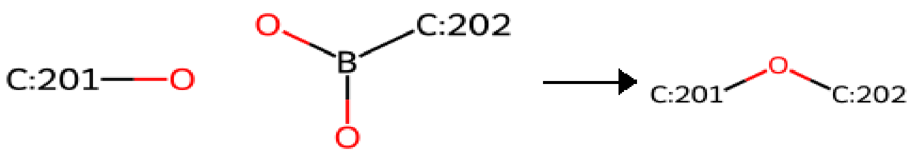
chan-lam-1

Chan-Lam coupling: N-arylation of anilines with boronic acids.



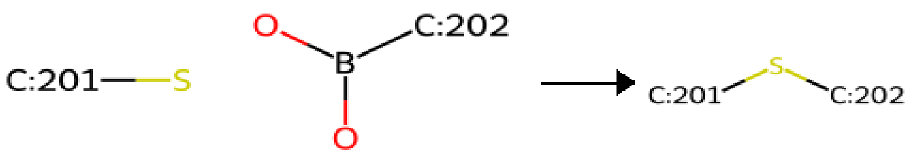
chan-lam-2

Chan-Lam coupling: O-arylation of phenols with boronic acids.



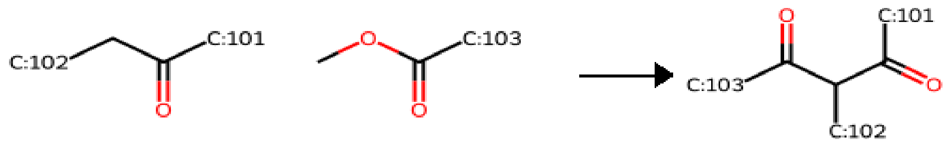
chan-lam-3

Chan-Lam coupling: S-arylation of thiophenols with boronic acids.



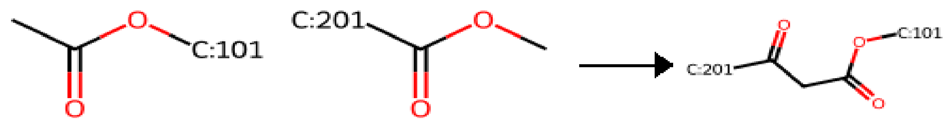
claisen-1

Crossed Claisen condensation: 1,3-diketone synthesis from an enolizable ketone and a non-enolizable ester.



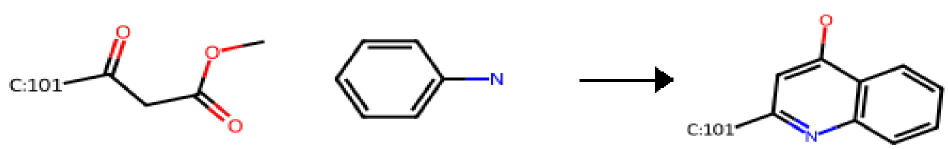
claisen-2

Claisen condensation: beta-keto ester synthesis from an enolizable ester and a non-enolizable ester.



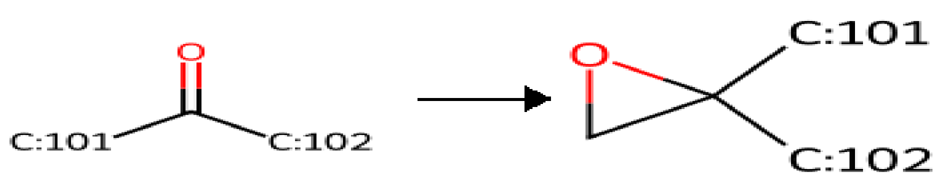
conrad-limpach

Conrad-Limpach synthesis of substituted 4-hydroxyquinolines.



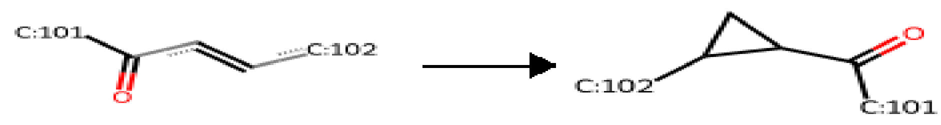
corey-chaykovsky-1

Corey-Chaykovsky synthesis of an epoxide from an aldehyde or a ketone.



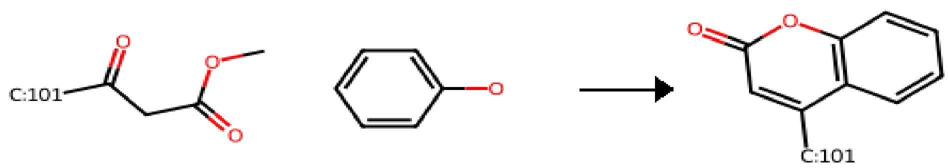
corey-chaykovsky-2

Corey-Chaykovsky cyclopropane synthesis.



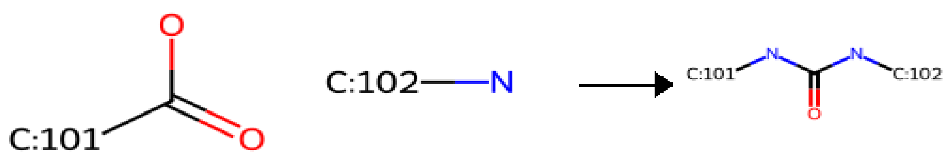
coumarin

Pechmann condensation: Coumarin synthesis from a phenol and a beta-keto ester.



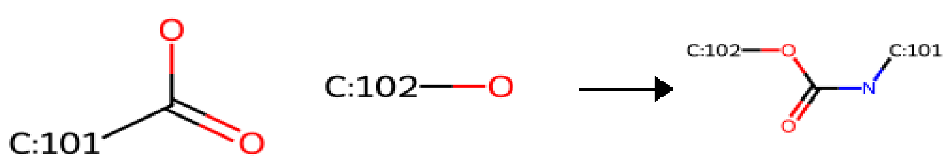
curtius-1

Curtius rearrangement: ureas via isocyanates formed from carboxylates



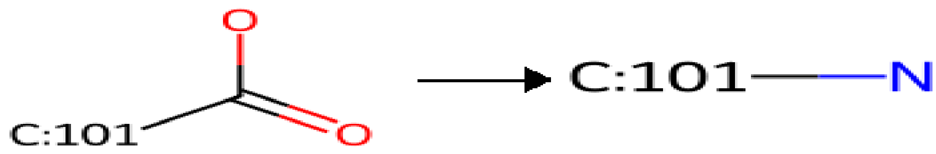
curtius-2

Curtius rearrangement: carbamates via isocyanates formed from carboxylates



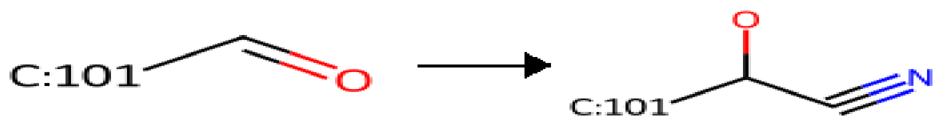
curtius-3

Curtius rearrangement: formation of primary amines via hydrolysis of isocyanates formed from carboxylates



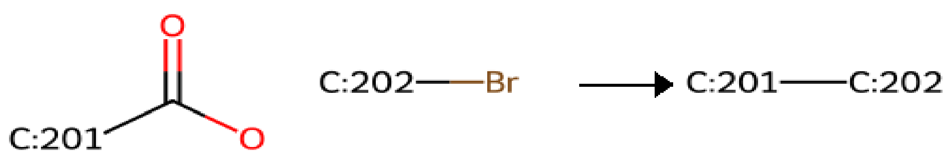
cyanohydrin

Cyanohydrin formation from an aldehyde.



decarboxylative_coupling-1

Aryl C-C bond formation from an aryl carboxylic acid and an aryl halide.

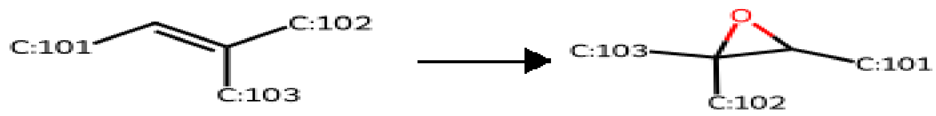


diyne

Heterocoupling of two alkynes to form a 1,3-diyne (implicit formation of a haloalkyne).

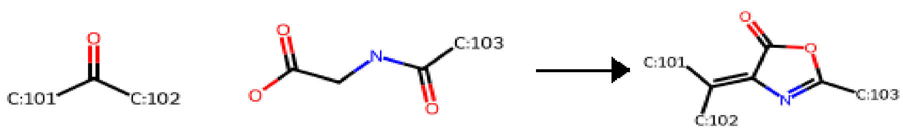
epoxidation

Epoxidation of alkenes.



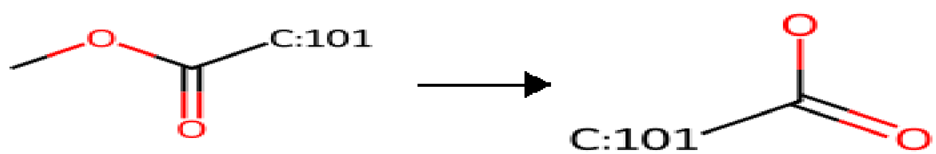
erlenmeyer-plochl

Erlenmeyer-Plochl azlactone synthesis from aldehydes or ketones and N-acyl glycine.



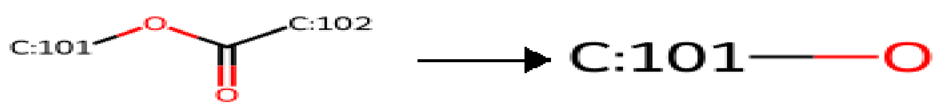
ester-hydrolysis-1

Hydrolysis of an ester to the parent carboxylic acid.



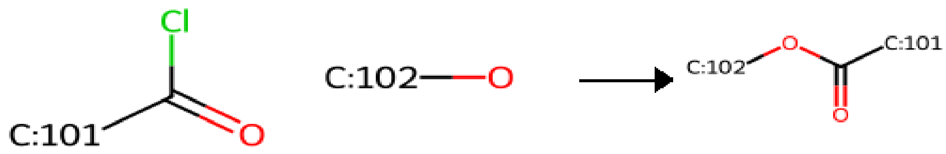
ester-hydrolysis-2

Hydrolysis of an ester to the parent alcohol.



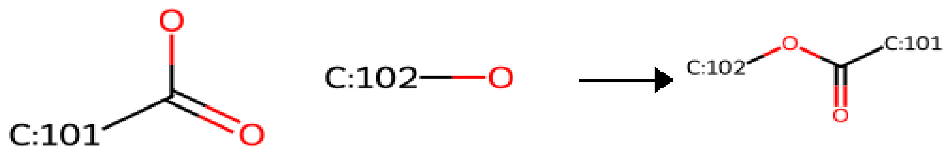
esterification-1

Esterification of an alcohol using an acid chloride.



esterification-2

Esterification of an alcohol and a carboxylic acid.



ether-1

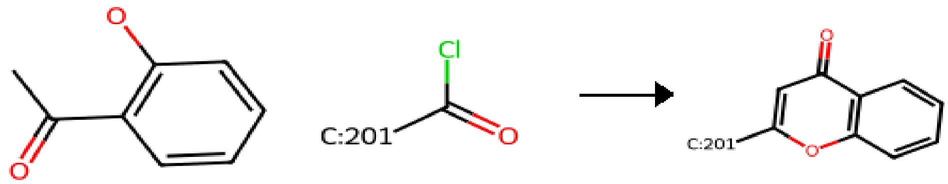
Ullmann aryl ether synthesis: gives a biaryl ether from an aryl halide and a phenol.

ether-2

Williamson ether synthesis: gives an ether from a primary or secondary halide and a primary, secondary or (hetero)aryl alcohol.

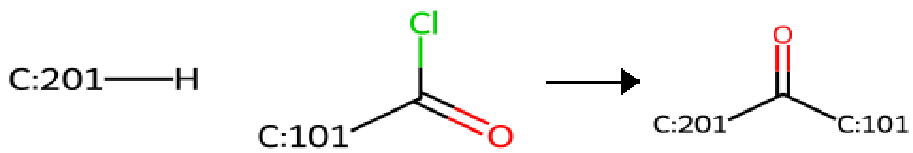
flavone

Flavone synthesis: gives a 2-aryl flavone from a 2-acyl phenol and an aryl acid chloride.



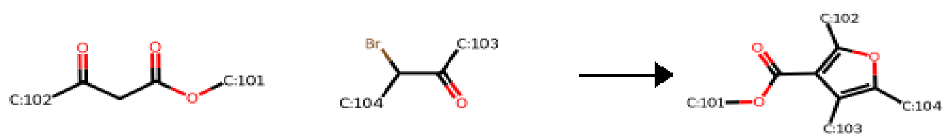
friedel-crafts-1

Friedel-Crafts acylation.



furan-1

Feist-Benary furan synthesis from a beta-keto-ester and a 2-halo ketone.

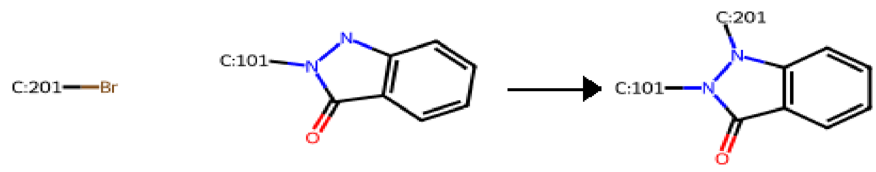


goldberg-1

Goldberg reaction: Ullman condensation of an aryl halide with an aniline or acetanilide.

goldberg-2

Goldberg reaction: Ullman condensation of an aryl halide with a benzylpyrazolone.

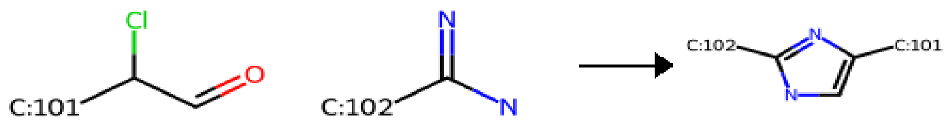


heck

Heck reaction: coupling of an activated alkene and aryl or vinyl halides.

imidazole-1

Imidazole synthesis from an alpha-chloro aldehyde and an amidine.

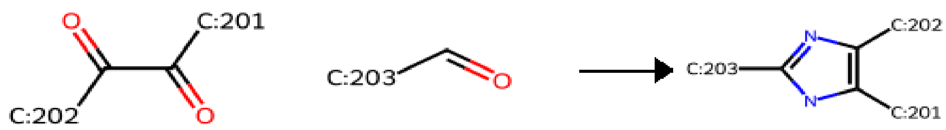


imidazole-2

Imidazole synthesis from an aryl aldehyde, 2-halo ketone and a primary amine.

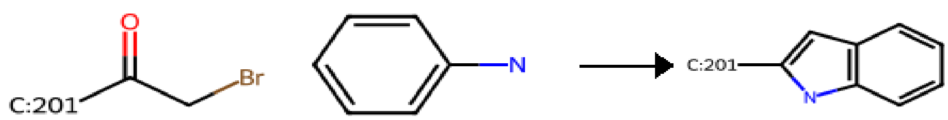
imidazole-3

Synthesis of triaryl-imidazoles.



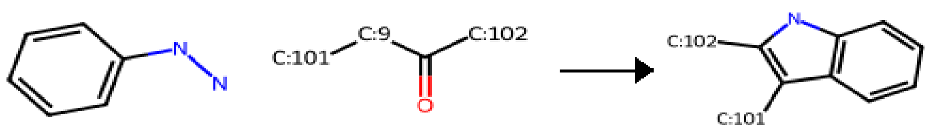
indole-1

Bischler-Mohlau indole synthesis: gives a 2-aryl Indole from an aniline and a 2-halo aryl ketone.



indole-2

Fischer indole: gives an indole from an aryl hydrazine and a ketone.

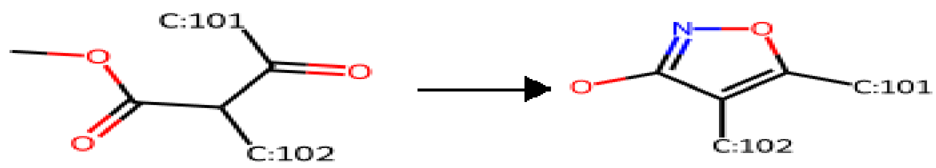


isocyanate-1

Isocyanate formation from primary amines.

isoxazole-1

Claisen isoxazole synthesis from a beta-ketoester and hydroxylamine.

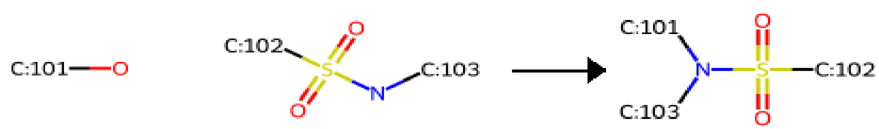


mitsunobu-1

Mitsunobu: gives an aryl ether from an alcohol and a phenol.

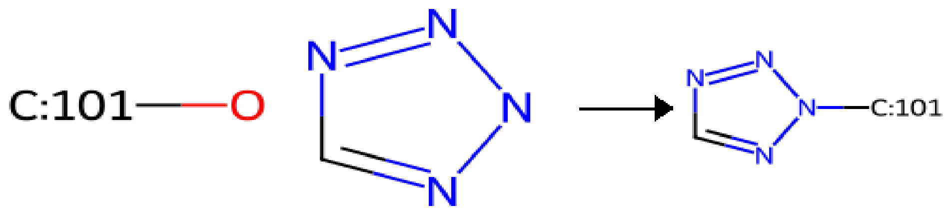
mitsunobu-2

Mitsunobu: alkylation of a sulfonamide with a primary or secondary alcohol.



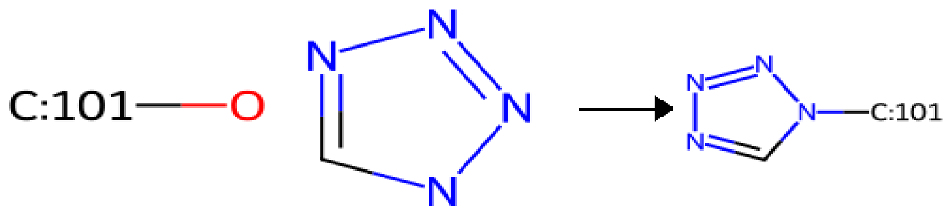
mitsunobu-3

Mitsunobu: N-alkylation of a tetrazole with a primary or secondary alcohol.



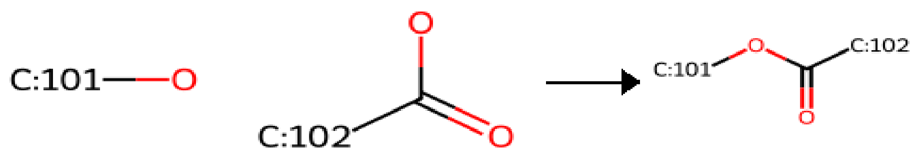
mitsunobu-4

Mitsunobu: N-alkylation of a tetrazole with a primary or secondary alcohol.



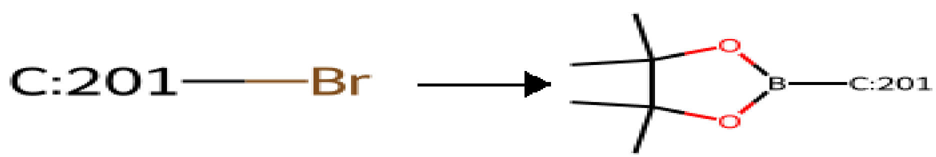
mitsunobu-5

Mitsunobu: esterification from an alcohol and a carboxylic acid.



miyaura-borylation

Miyaura borylation.

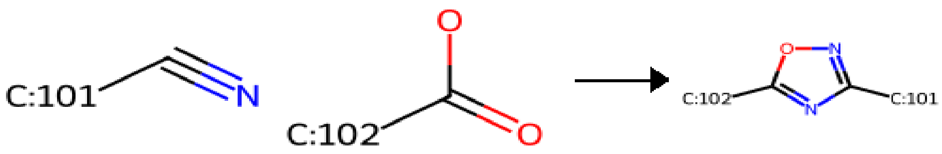


negishi

Negishi: aryl C-C coupling from two halides (one will be prepared as an organozinc).

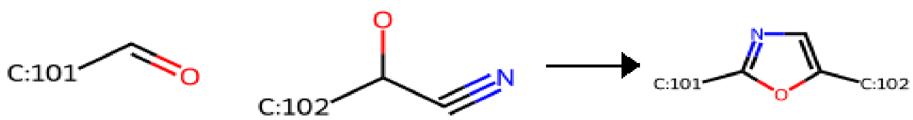
oxadiazole-1

Oxadiazole synthesis from a nitrile (converted to an amidoxime) and a carboxylic acid.



oxazole-1

Fischer Oxazole Synthesis: Generates an oxazole from an aldehyde and a cyanohydrin.

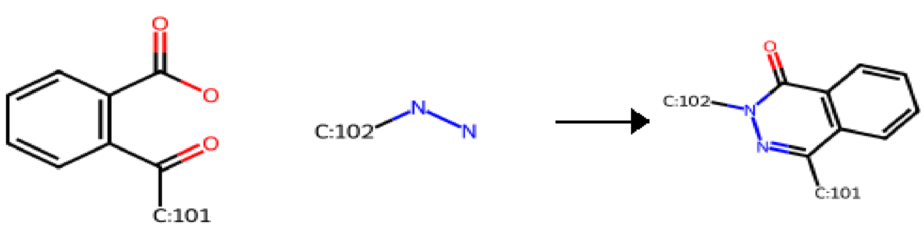


oxazole-2

2,4-Disubstituted oxazole synthesis from monosubstituted alkynes and acetonitrile or proprionitrile (PhIO as oxygen source).

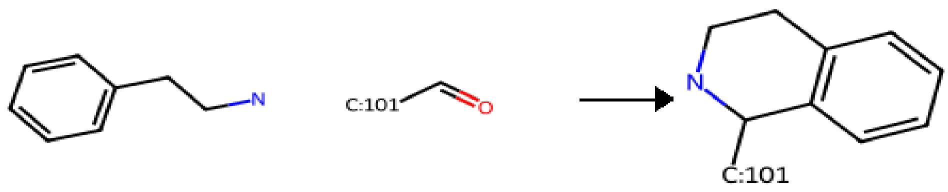
phthalazinone

Phthalazinone synthesis from 2-acylbenzoic acid and a hydrazine.



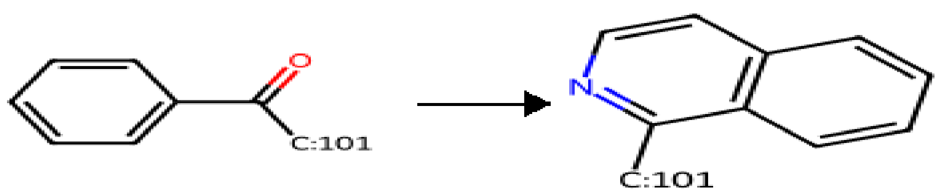
pictet-spengler

Pictet-Spengler reaction.



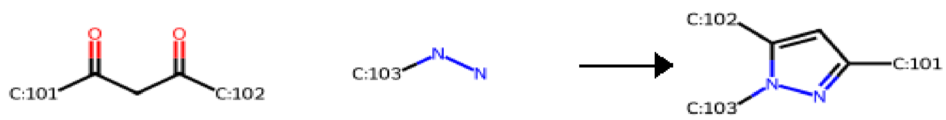
pomeranz-fritsch

Pomeranz-Fritsch synthesis of isoquinolines.



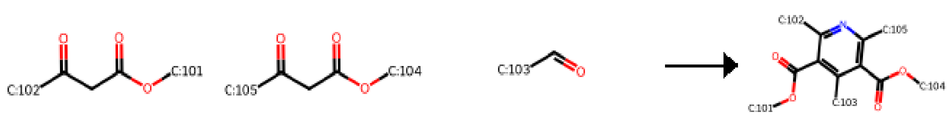
pyrazole-1

Pyrazole synthesis from 1,3-diketones and hydrazines.



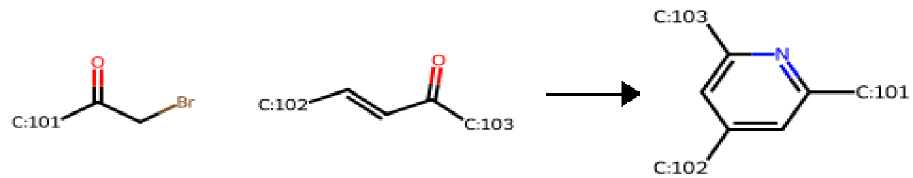
pyridine-1

Hantzsch pyridine synthesis from two beta-ketoesters, an aldehyde, and ammonia (stepwise).



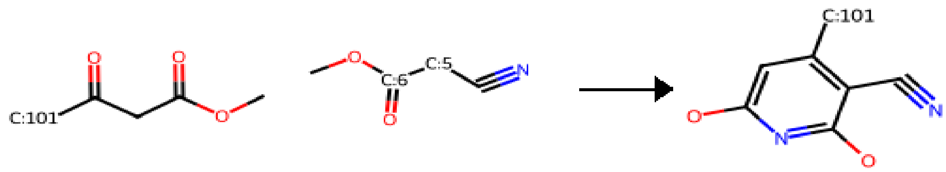
pyridine-2

Krohnke pyridine synthesis from a 2-halo ketone and an alfa,beta-unsaturated ketone.



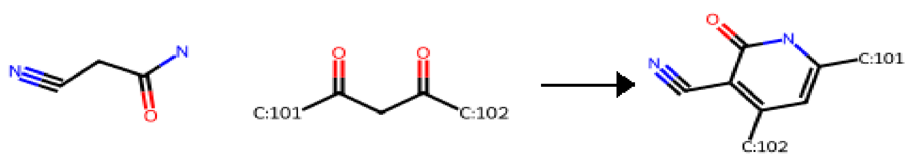
pyridine-3

Guareshi-Thorpe pyridine synthesis from a beta-ketoester, and a cyanoacetic ester.



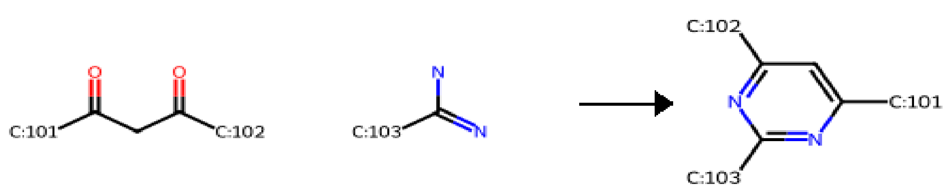
pyridone-1

Pyridone synthesis from cyanoacetamide and 1,3-diketones.



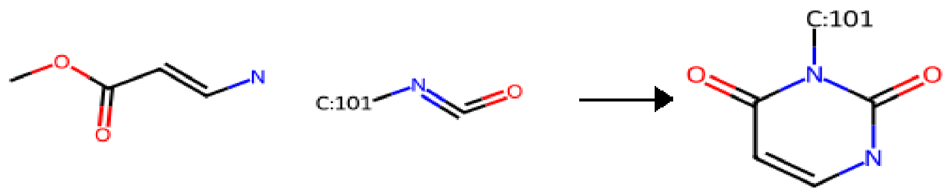
pyrimidine-1

Pinner pyrimidine synthesis.



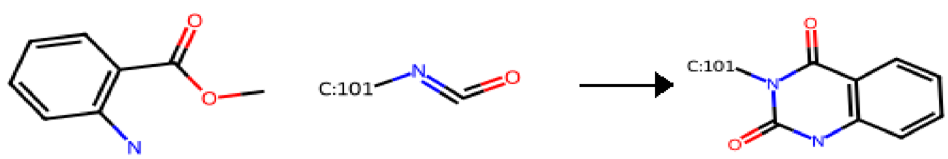
pyrimidinedione-1

Pyrimidinedione synthesis from amino-acrylates and isocyanates.



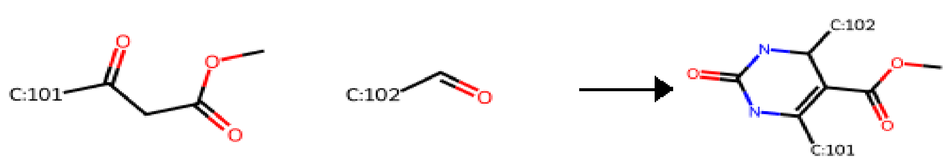
pyrimidinedione-2

Bicyclic pyrimidinedione synthesis from 2-aminobenzoates and isocyanates.



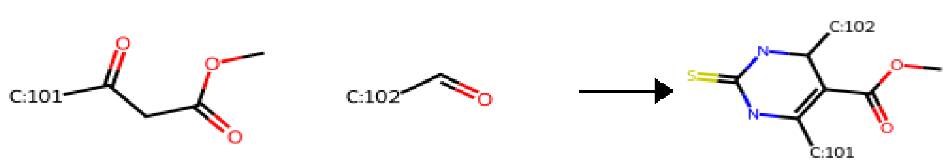
pyrimidone-1

Biginelli 2-pyrimidone synthesis.



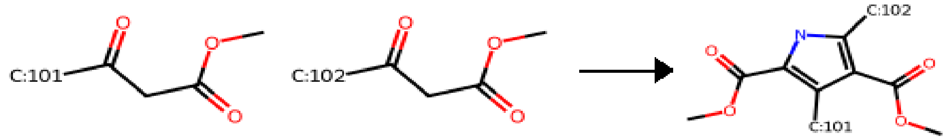
pyrimidone-2

Biginelli 2-mercaptopyrimidine synthesis.



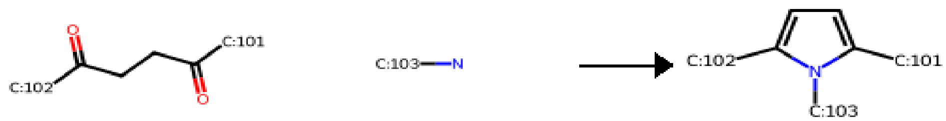
pyrrole-1

Knorr pyrrole synthesis.



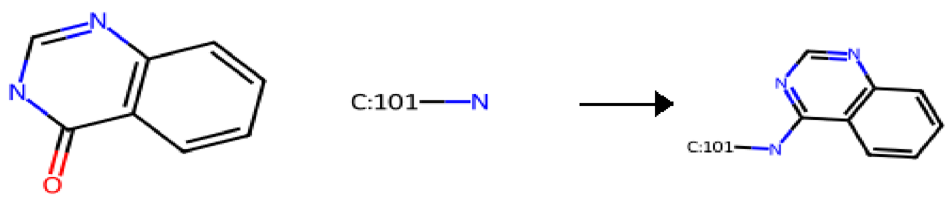
pyrrole-2

Paal-Knorr pyrrole synthesis.



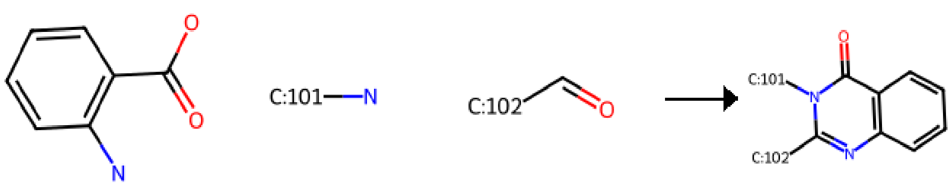
quinazoline-1

Quinazoline synthesis from a quinazolinone and a primary or secondary amine.



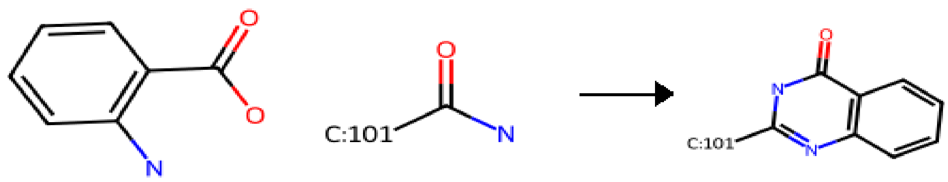
quinazolinone-1

Quinazolinone synthesis from a 2-carboxy aniline, a primary amine, and an aldehyde.



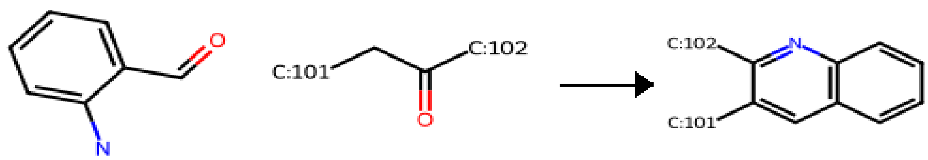
quinazolinone-2

Niementowski quinazolinone synthesis from a 2-carboxy aniline, and a primary amide.



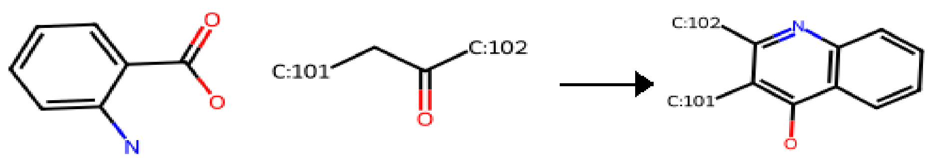
quinoline-1

Quinoline synthesis from a 2-amino benzaldehyde and a ketone.



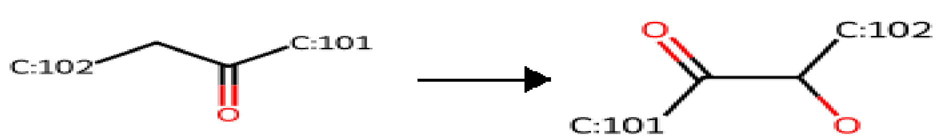
quinoline-2

Niementowski quinoline synthesis from a 2-carboxy aniline and a ketone.



rubottom

Rubottom oxidation: alfa hydroxylation of an enolizable ketone via the corresponding silyl enol ether.

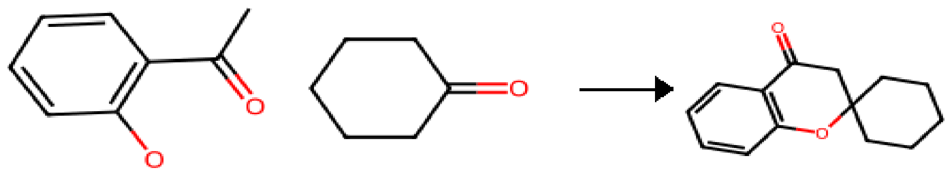


sonogashira

Sonogashira Coupling: Arylation of an alkyne with an aryl/vinyl halide.

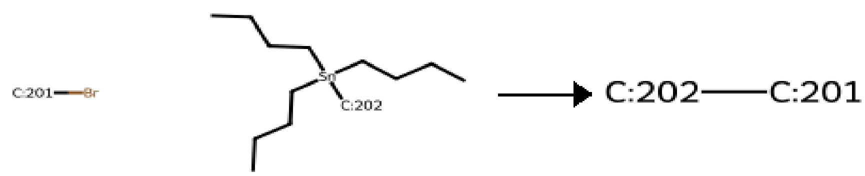
spiro-chromanone

Spiro-chromanone synthesis from an o-hydroxyacetophenone and a cyclohexanone.



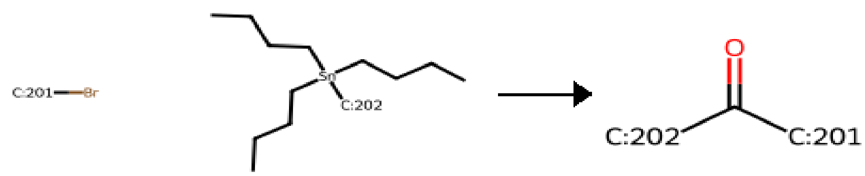
stille

Stille Coupling: aryl C-C coupling from an aryl or vinyl halide and an organostannane.



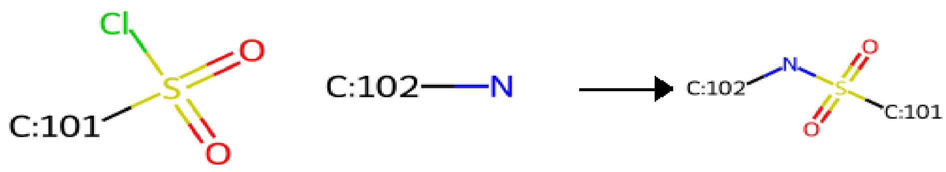
stille-2

Stille Carbonylative Coupling: aryl C-C coupling with CO insertion from an aryl or vinyl halide and an organostannane.



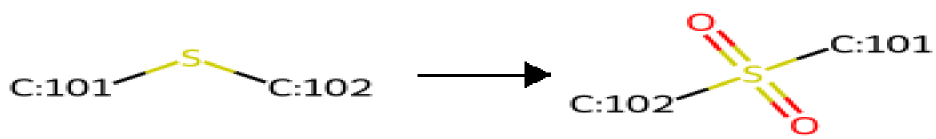
sulfonamide-1

Sulfonamide synthesis from a sulfonyl chloride and a primary or secondary amine.



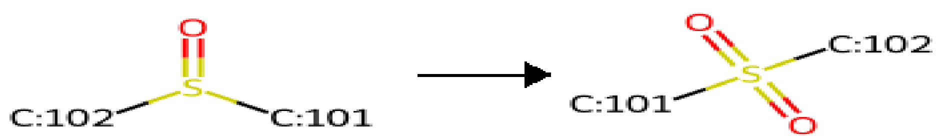
sulfone-1

Oxidation of a thioether to a sulfone.



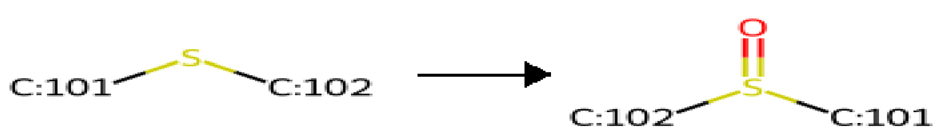
sulfone-2

Oxidation of a sulfoxide to a sulfone.



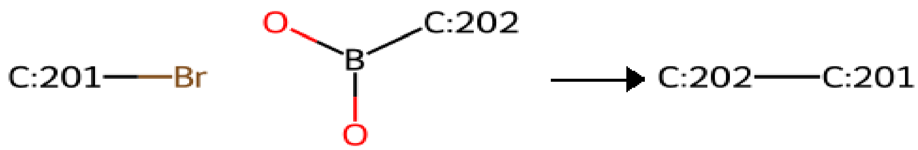
sulfoxide

Oxidation of a thioether to a sulfoxide.



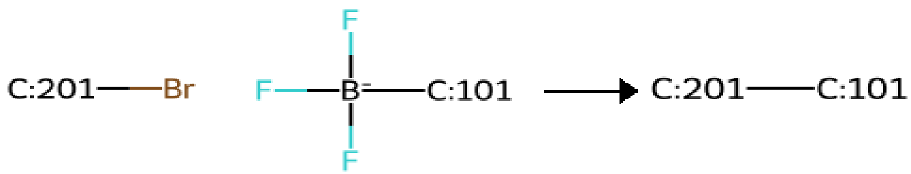
suzuki

Suzuki coupling: aryl C-C coupling from an aryl halide and an aryl/vinyl boronate.



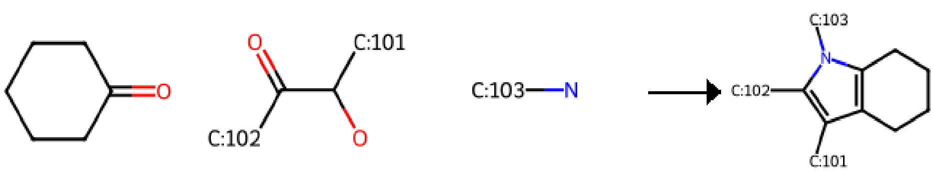
suzuki-2

Suzuki coupling: C-C coupling of an aryl halide and a trifluoroborate.



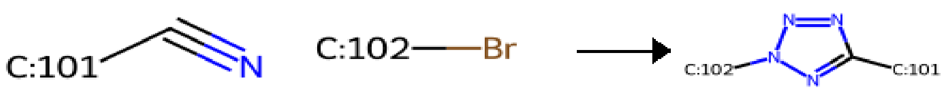
tetrahydroindole-1

Tetrahydroindole synthesis from a cyclohexanone, 2-OH ketone and a primary amine.



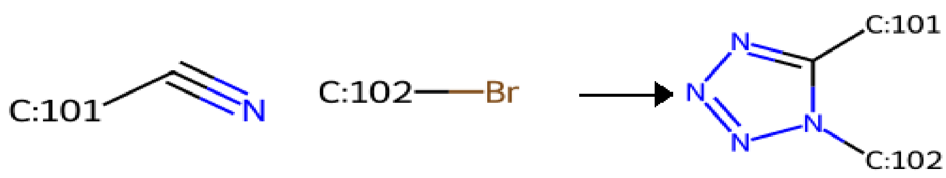
tetrazole-1

Tetrazole from nitrile and alkyl halide; first regioisomer.



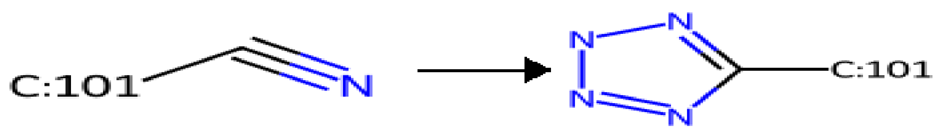
tetrazole-2

Tetrazole from nitrile and alkyl halide; second regioisomer.



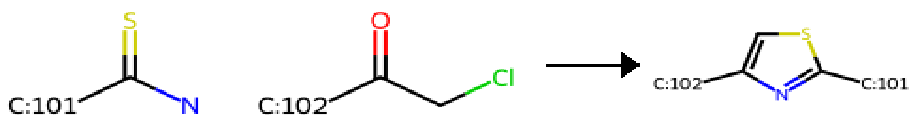
tetrazole-3

Synthesis of a terminal tetrazole from a nitrile.



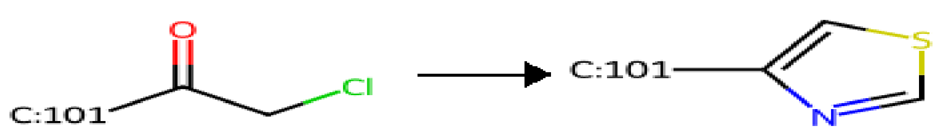
thiazole-1

Hantzsch thiazole synthesis from a thioamide and alpha-halo ketone.



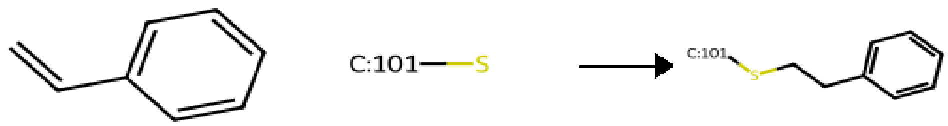
thiazole-2

Hantzsch thiazole synthesis from thioformamide and an alpha-halo ketone.



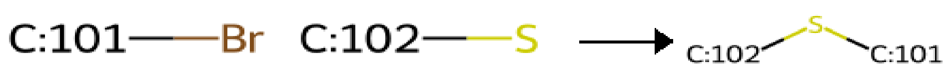
thioether-1

Thioether synthesis from styrenes and thiols.



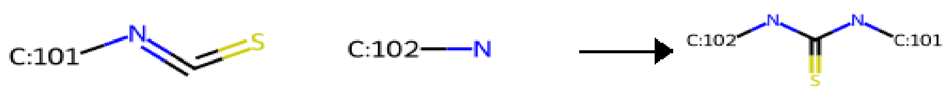
thioether-2

Thioether synthesis from halides and thiols.



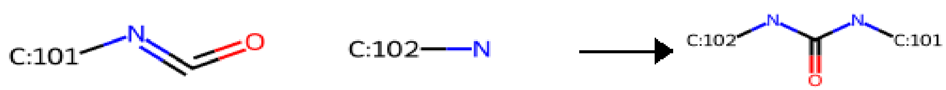
thiourea-1

Thiourea synthesis from thioisocyanates and primary or secondary amines.



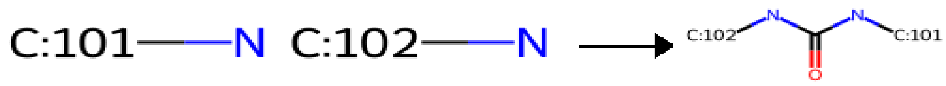
urea-1

Urea synthesis from isocyanates and primary or secondary amines.



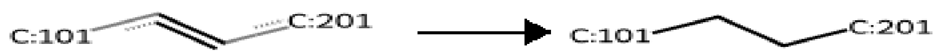
urea-2

Urea synthesis from primary or secondary amines and primary or secondary amines.



vinyl-reduction

Reduction of vinyl groups.



yamaguchi

Yamaguchi macrolactonization.